

HEMA PRIYA PRAKASH

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EDUCATION

UNIVERSITY OF ARIZONA

2023 - 2025

MS, Information Science

- **Coursework:** Neural Networks, Data Analytics and Visualization, Data Mining and Discovery, SQL/NoSQL for Database Design, Applied NLP, Data Warehousing and Analytics, AI/ML

GPA: **3.5/4.0**

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

2018 - 2022

Bachelor of Engineering, Computer Science

GPA: **3.5/4.0**

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, HTML/CSS, C, C++

ETL Tools: Informatica Power center, Apache Airflow, Azure ADF, Control-M, Hadoop and Pyspark, Docker, Kubernetes

Cloud: Snowflake, IICS

Analytics Tools: PowerBI, Tableau, AWS (S3, Lambda, Redshift)

Databases: MySQL, MongoDB, SQLite

Analytical & Statistical Techniques: Data Mining, Predictive Modelling, Pattern Recognition, Kalman Filters, RAG tools

Theoretical Concepts: Data warehousing and Data Modeling, Machine Learning concepts, Data Science

ML Development & Concepts: LLM, Deep Learning, Timeseries Forecasting, Hyperparameter Tuning, Torch Vision, NLP

Frameworks and Libraries: NumPy, Pandas, Scikit-Learn, Seaborn, Matplotlib, SciPy, Keras, TensorFlow, NLTK

EXPERIENCE

- Hands-on experience in **data analysis, machine learning, and SQL/NoSQL database management** through academic and capstone projects.
- Designed a **Travel Booking System database** (conceptual, logical, physical models) to improve data consistency and reporting.
- Contributed to the **UA Navigation & Accessibility Capstone Project**, building a **Python-based AI text navigation assistant** and enhancing accessibility through UI/UX design in Figma.
- Collaborated in a **cross-functional team** to conduct system testing, debugging, and data visualization, strengthening teamwork, communication, and problem-solving skills.
- Developed strong proficiency in **Excel, data cleaning, and documentation** while supporting project deliverables and quality assurance tasks.

ACADEMIC PROJECTS

AI NAVIGATION ASSISTANT - CAPSTONE PROJECT | UNIVERSITY OF ARIZONA | 2025

- Collaborated in a 4-member team to design an **accessibility-first navigation system**.
- Supported **backend integration using FastAPI** and contributed to the development of an **AI-powered text-based navigation assistant** leveraging **computer vision, OCR, and NLP techniques**.
- Managed documentation, updates, and task tracking to ensure project alignment.

TRAVEL BOOKING SYSTEM – DATABASE PROJECT

- Designed schemas, ER diagrams, and documentation for a **booking system database**.
- Wrote and executed **SQL queries** (joins, subqueries, triggers) and **MongoDB aggregation pipelines** to demonstrate both relational and NoSQL implementations.
- Assisted in project planning and ensured data accuracy.
- Collaborated with teammates to maintain **shared documentation** and streamline the database development process.

DATA ANALYSIS PROJECTS

- Built and maintained **Excel and Google Sheets dashboards** to track, visualize, and report on key dataset trends.
- Conducted **data wrangling, cleaning, and visualization** of public datasets using Python (Pandas, NumPy, Matplotlib).
- Documented findings and prepared **analytical reports** to support decision-making and communicate insights clearly.
- Applied basic statistical methods and exploratory data analysis (EDA) to identify trends, correlations, and actionable insights.

CERTIFICATIONS

IEEE Certified IoT (Internet of Things)

Coursera: Google Data Analytics Certificate

OpenAI Developer API (Projects)

SQL for Data Science – Coursera